LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration of construction and explanation on operation of Directional indicator, Artificial Horizon & TSI.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration of construction and explanation on operation of Directional indicator, Artificial Horizon & TSI.
5. Specific Objectives	24.5.1.1 Trainee will be able to explain the operation of Directional indicator, Artificial Horizon & TSI.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of construction and operation of Directional Indicator, Artificial Horizon, and TSI using L & Ds and IB; Safety: Ensure trainees maintain safe distance from rotating parts; Instructor supervision: Continuous monitoring during explanation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary flight instruments covered in this lesson?
(b) Why is the Artificial Horizon critical for instrument flight?
(c) What is the primary function of the Directional Indicator?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Construction of Directional Indicator • Operation of Directional Indicator • Sources of Error in Directional Indicator	L & Ds	PPT, IB	15 min
(b)	• Construction of Artificial Horizon • Operation of Artificial Horizon • Interpretation of Artificial Horizon	L & Ds	PPT, PDF	20 min
(c)	• Construction of Turn and Slip Indicator • Operation of Turn and Slip Indicator • Interpretation of Turn and Slip Indicator	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the gyroscope maintains stability in the Artificial Horizon.
(b) Describe the relationship between the ball position and aircraft slip in the TSI.
(c) How does the Directional Indicator correct for magnetic variation?

12. Home Assignment and Follow Up	Prepare a brief note on the limitations of the Directional Indicator and how it relates to the next lesson on HEIU.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of construction and explanation on operation of HEIU. Safety precautions to be observed while working on HEIU.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min